18/March/26 * EXPERIMENT- 08 *

* **AIM** :- Determining the probability of a given case using binomial distribution.

* **EXPERIMENT** :- Ten coins are thrown simultaneously. Find the probability of getting at least seven heads.

* **THEORY** :- A random variable X has the binomial distribution if its probability distribution is given by ;

$$\Rightarrow \boxed{P[X = x] = {}^{n}C_{x}\, p^{x} q^{n-x}}$$

$$, \quad x = 0, 1, 2, 3, ---, n$$

* **RESULT** :-

→ The probability of getting at least seven heads after throwing 10 coins simultaneously is ;

$$120\left(\tfrac{1}{2}\right)^{10} + 45\left(\tfrac{1}{2}\right)^{10} + 10\left(\tfrac{1}{2}\right)^{10} + \left(\tfrac{1}{2}\right)^{10}$$

—— * —— * —— * —— *

CALCULATIONS

Let p be the probability of getting a head,

$$p = 1/2$$

and q be the probability of getting a failure,

$$q = 1 - p = 1/2$$

Using binomial distribution,
The probability of getting x heads in a throw of 10 coins is,

$$p(x) = {}^{10}C_x \left(\tfrac{1}{2}\right)^x \left(\tfrac{1}{2}\right)^{10-x}$$

$$= {}^{10}C_x \left(\tfrac{1}{2}\right)^{10}, \quad x = 0, 1, 2 \ldots, 10$$

Now, probability of getting atleast 7 heads is given by;

$$P(X \geq 7) = P(7) + p(8) + p(9) + p(10)$$

$$= {}^{10}C_7 \left(\tfrac{1}{2}\right)^{10} + {}^{10}C_8 \left(\tfrac{1}{2}\right)^{10} + {}^{10}C_9 \left(\tfrac{1}{2}\right)^{10} + {}^{10}C_{10} \left(\tfrac{1}{2}\right)^{10}$$

$$\therefore \; 120\left(\tfrac{1}{2}\right)^{10} + 45\left(\tfrac{1}{2}\right)^{10} + 10\left(\tfrac{1}{2}\right)^{10} + \left(\tfrac{1}{2}\right)^{10}$$